



Enhancing multimodal deepfake detection with local–global feature integration and diffusion models

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Abstract

Deepfake detection has become a critical challenge nowadays with the rise of sophisticated generative techniques that manipulate audio-visual data. Existing methods primarily focus on lip movement synchronization using audio and visual features, often relying on local feature extraction with Convolutional Neural Networks (CNNs). In this work, we propose an enhanced multimodal framework that integrates with local and global features for advanced deepfake detection. Our approach extends traditional pipelines by introducing additional visual features such as eye movement and facial regions, combined with audio features to model cross-modal dependencies. While CNNs capture local features, Vision Transformers (ViTs) extract global contextual relationships from both visual and audio modalities. The diffusion models are incorporated as pre-processors to refine noisy data and generate realistic augmentations, ensuring high-quality feature representation. The proposed framework achieves state-of-the-art performance, with accuracy scores of 0.9987, 0.9825, 0.9915, and 0.9812 on the FakeAVCeleb, AV-Deepfake1M, TVIL, and LAV-DF datasets, respectively. These results demonstrate significant improvements over existing methods, highlighting the framework's superior generalization and robustness in detecting subtle inconsistencies across manipulated audio-visual data.

Keywords Audio-visual synchronization · Deepfake detection · Diffusion models · Multimodal analysis · Vision transformers

1 Introduction

The rise of deepfake technology has led to increasing concerns about the authenticity of media. Deepfakes-manipulated videos, images and audio are generated using artificial intel-

ligence (AI) to create highly realistic but fake content. While these techniques can be used for entertainment and creative purposes, they also present significant risks, especially in terms of misinformation, privacy violations, and cyber threats [1, 2]. The ability to convincingly alter both video and audio has made it increasingly difficult to distinguish between real and fabricated media [3].

Existing deepfake detection methods typically focus on either visual or audio features independently, but this approach is limited when both modalities are altered [3–7]. For example, in cases where the video is real but the audio is fake, or vice versa, detection becomes more challenging [8, 9]. To address this, we propose a multimodal framework that analyzes both audio and visual features in tandem. By combining these modalities, our approach can better capture inconsistencies across both video and audio, improving detection accuracy.

In this work, we propose a novel framework for deepfake detection that goes beyond traditional methods by incorporating both local and global features from audio and visual data. Alongside lip synchronization, we also focus on additional visual features such as eye movements and the overall

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facial region [1, 10]. We use Convolutional Neural Networks (CNNs) for extracting local features (like lips and eyes) and Vision Transformers (ViTs) for capturing broader, global patterns in the data, providing a more holistic view of the content.

Additionally, we incorporate Diffusion Models to improve the quality of both the audio and visual data. These models help to reduce noise and enhance features, ensuring that our system works effectively even when the input data is of lower quality.

The main contributions of this paper are:

- We add new visual features, such as eye movements and facial regions, to complement lip synchronization, improving the detection of deepfakes.
- We combine local feature extraction (via CNNs) and global context modeling (via ViTs) to create a more accurate and robust detection system.
- We use Diffusion Models to refine the input data, improving feature quality and boosting the system's resistance to noise and manipulation

We evaluate our framework on several well-known deepfake detection datasets, including FakeAVCeleb [11], AV-Deepfake1M [12], TVIL [13], and LAV-DF [14]. Our results show that the proposed method outperforms existing approaches, offering higher accuracy, precision, and recall while ensuring better generalization across different manipulation techniques and datasets.

The paper is organized as follows: Sect. 2 reviews the related work in deepfake detection. Section 3 describes the materials and methods used in our framework. Section 4 presents the experimental results, followed by a discussion comparing our approach with previous models. Finally, Sect. 5 concludes the paper and discusses potential future directions.

2 Related work

The growing challenge of detecting deepfake content has led to extensive research on various detection methods, focusing on both audio-visual synchronization and multimodal fusion approaches. Early methods in deepfake detection relied primarily on analyzing individual modalities, such as video or audio, to identify inconsistencies or artifacts [15]. However, as deepfake technology has advanced, more sophisticated techniques have emerged that combine multiple modalities, such as both visual and auditory features, to achieve higher accuracy in detection [15].

2.1 Multimodal deepfake detection

Multimodal deepfake detection approaches have gained attention due to their potential to enhance accuracy by analyzing both visual and audio features simultaneously. A notable work in this area introduced an audio-visual synchronization framework to detect inconsistencies between lip movements and audio features. This approach uses Convolutional Neural Networks (CNNs) and Temporal Convolutional Networks (TCNs) for processing visual and audio data, respectively, and employs Bidirectional Long Short-Term Memory (BiLSTM) [16] networks combined with Dynamic Time Warping (DTW) to assess synchronization [17]. However, these methods primarily focus on late fusion strategies, which may fail to capture complex temporal dependencies between the modalities, limiting their effectiveness in certain deepfake scenarios [17].

Further research has extended this approach by incorporating various fusion techniques, including hybrid models and ensemble methods. For instance, some studies have integrated CNNs and RNN-based architectures to capture both spatial and temporal features [18–20]. While these methods offer improvements in detection performance, they still face challenges in handling real-time applications and ensuring robustness across different types of deepfake manipulations [21].

2.2 Deepfake detection via feature fusion

Several other studies have explored feature fusion strategies to combine the strengths of different feature extraction methods. For example, some researchers have proposed combining CNN features for visual processing with recurrent models like BiLSTM to capture temporal dynamics in the audio [14, 22, 23]. In this context, deep learning models that focus on fine-grained analysis of lip synchronization have shown promising results [21]. However, issues related to computational efficiency and real-time processing remain significant hurdles in the adoption of these methods in practical applications [24].

2.3 Innovations in audio-visual synchronization

One significant contribution to deepfake detection involves leveraging the natural synchronization between audio and visual signals. The work of [17] explored the use of multimodal learning techniques to model inconsistencies between the video and audio. This method primarily detects manipulation by identifying misalignments in audio-visual synchronization, providing an important step forward in distinguishing real content from fabricated media [18, 19]. However, integrating these features with high accuracy and in real-time systems remains a complex task [20, 21, 25].

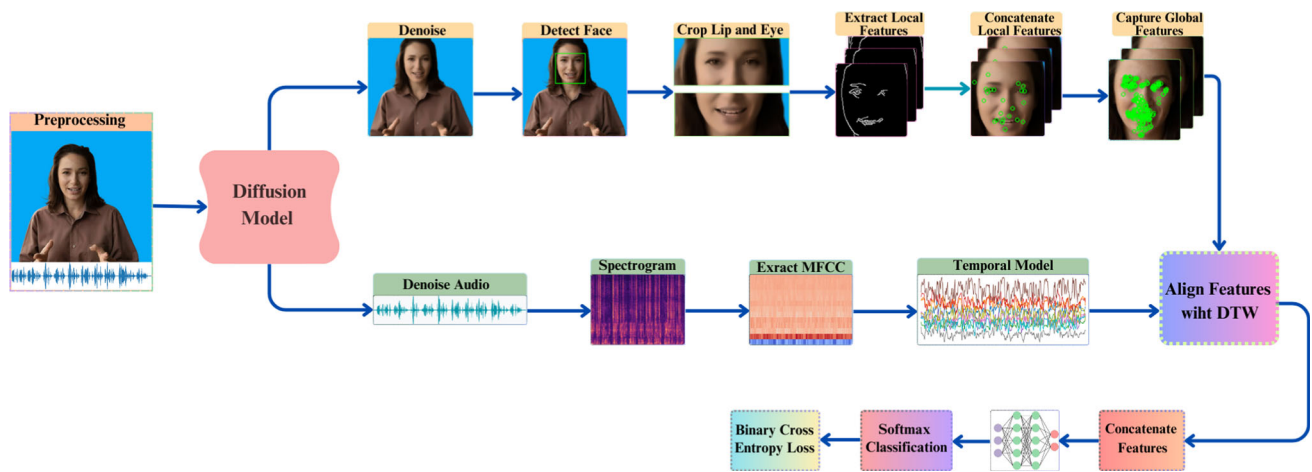


Fig. 1 A system architecture of the proposed framework to achieve advanced and accurate deepfake detection

2.4 Advancements in deepfake generation techniques

Recent studies have also highlighted various techniques for generating deepfakes, such as face synthesis and expression swapping, which have furthered the complexity of detection tasks [21–23]. The ability to swap faces, alter expressions, and manipulate audio tracks independently of video has led to significant challenges in deepfake detection [14]. Some efforts have attempted to address this by using hybrid models, combining traditional face recognition techniques with advanced deep learning models [26, 27]. Despite these advancements, detecting deepfakes involving subtle alterations, like mismatched lip-sync or fine-grained audio manipulations, still poses a major challenge [28, 29].

2.5 Recent detection frameworks

Several modern deepfake detection frameworks focus on multimodal fusion, leveraging both video and audio features to detect inconsistencies in real-time applications. For instance, the study presented by [18, 30] integrates both video frames and audio features through deep neural networks, providing promising results on datasets like FaceForensics++ and CelebDF. However, the need for large datasets and the computational demands of these methods can hinder their practical deployment [31, 32].

In contrast, our proposed approach focuses on a fine-grained analysis of lip synchronization by combining visual features, such as eye movements and facial regions, with audio features. By using Vision Transformers (ViTs) for global feature extraction and Diffusion Models to refine both audio and visual inputs, our framework aims to improve detection accuracy while ensuring computational efficiency. We evaluate our model on multimodal datasets such as

FakeAVCeleb [11], AV-Deepfake1M [12], TVIL [13], and LAV-DF [14], which provide diverse and challenging data for detecting complex deepfake manipulations.

These work highlights the rapid progress in deepfake detection techniques, particularly through multimodal analysis. While significant advancements have been made, challenges remain in ensuring real-time, robust, and scalable detection systems. Our approach aims to build on these advancements by focusing on precise lip-sync analysis, integrating both visual and audio features, and refining data through Diffusion Models for improved accuracy and efficiency in deepfake detection.

3 Methodology

The proposed framework integrates local and global feature extraction through Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to capture fine-grained and contextual relationships across visual and audio modalities Fig. 1. Additionally, diffusion models preprocess noisy data, enhancing the input quality. The methodology includes visual and audio feature extraction, multimodal fusion, classification, and preprocessing with diffusion models.

3.1 Visual feature extraction

Visual features are extracted using a two-stage pipeline: CNNs for local feature extraction and ViTs for global context modeling Fig. 2. For each input video frame F_t , the face region is detected:

$$F_{\text{face},t} = f(F_t) \quad (1)$$

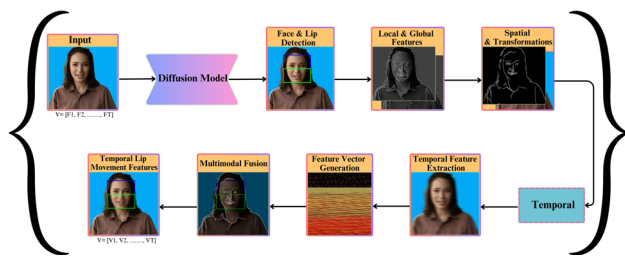


Fig. 2 Video processing architecture

where f is a face detection model and F_t is video frame. The lip and eye regions are cropped from $F_{\text{face},t}$:

$$F_{\text{lips},t} = l(F_{\text{face},t}), \quad F_{\text{eyes},t} = e(F_{\text{face},t}) \quad (2)$$

where l and e are lip and eye cropping functions.

The extracted regions are passed through CNNs to capture local features:

$$S_t^{\text{lips}} = \text{CNN}(F_{\text{lips},t}), \quad S_t^{\text{eyes}} = \text{CNN}(F_{\text{eyes},t}) \quad (3)$$

The outputs are concatenated:

$$S_t^{\text{local}} = [S_t^{\text{lips}}, S_t^{\text{eyes}}] \quad (4)$$

The sequence of local features $\{S_1^{\text{local}}, S_2^{\text{local}}, \dots, S_T^{\text{local}}\}$ is passed to a Vision Transformer (ViT) to capture global dependencies across frames:

$$H_t = \text{ViT}\left(\left\{S_1^{\text{local}}, S_2^{\text{local}}, \dots, S_T^{\text{local}}\right\}\right) \quad (5)$$

where H_t represents the global feature representation for frame t .

3.2 Audio feature extraction

Audio features are extracted using Mel-Frequency Cepstral Coefficients (MFCCs), which transform the raw audio signal $a(t)$ into frequency domain representations Fig. 3:

$$\text{MFCC}(a) = \mathcal{F}(X(t, f)) \quad (6)$$

where $X(t, f)$ is the spectrogram obtained from the Short-Time Fourier Transform (STFT) [33]:

$$X(t, f) = \sum_{n=0}^{N-1} a[n]e^{-i2\pi fn} \quad (7)$$

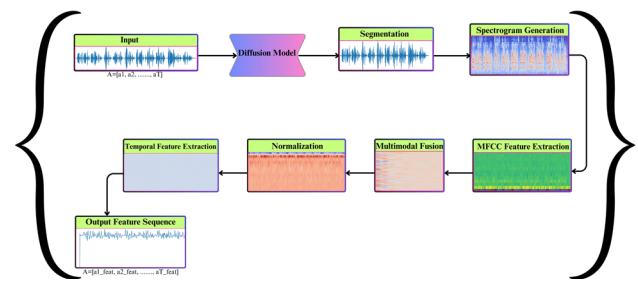


Fig. 3 Audio processing architecture

The MFCC features are modeled using a BiLSTM to capture temporal dependencies:

$$h_t^{\text{audio}} = \text{BiLSTM}(\text{MFCC}(a_t)) \quad (8)$$

resulting in the audio feature sequence:

$$\mathbf{a} = \{h_1^{\text{audio}}, h_2^{\text{audio}}, \dots, h_T^{\text{audio}}\} \quad (9)$$

3.3 Multimodal fusion

The visual and audio features are aligned using Dynamic Time Warping (DTW) to ensure temporal synchronization:

$$\text{DTW}(\mathbf{v}, \mathbf{a}) = \min \sum_{t=1}^T |v_t - a_t| \quad (10)$$

where $\mathbf{v}$ and $\mathbf{a}$ are visual and audio feature sequences. The aligned features are concatenated:

$$F_t = [v_t^{\text{aligned}}, a_t^{\text{aligned}}] \quad (11)$$

3.4 Classification

The fused features are passed through a fully connected neural network:

$$\hat{y} = \text{Softmax}(\mathbf{W}F_t + b) \quad (12)$$

where $\mathbf{W}$ and b are the learnable weights and biases. A binary cross-entropy loss function is used for optimization:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right] \quad (13)$$

3.5 Preprocessing with diffusion models

To enhance the quality of both visual and audio inputs, we employ Denoising Diffusion Probabilistic Models (DDPMs) as a preprocessing step. DDPMs iteratively refine noisy data into high-quality features, making them particularly effective for handling low-quality or artifact-laden inputs.

For visual data, the DDPM refines noisy frames by reversing a stochastic noise addition process:

$$q(x_{t-1}|x_t) = \mathcal{N}(\sqrt{\alpha_t}x_t, (1 - \alpha_t)\mathbf{I}) \quad (14)$$

where x_t is the noisy frame at time step t , α_t is a predefined noise schedule, and $\mathcal{N}$ denotes a Gaussian distribution.

The denoised frame x_0 is recovered through a learned reverse process:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (15)$$

where μ_θ and Σ_θ are the predicted mean and variance parameters of the denoising step, learned via a neural network.

For audio, the DDPM operates similarly, enhancing the clarity of noisy spectrograms. Given an audio input $a(t)$, the diffusion model progressively reconstructs the original signal:

$$a_0 = \mathcal{D}(a_t; \theta) \quad (16)$$

where a_t is the noisy spectrogram, $\mathcal{D}$ is the learned denoising process, and θ are the model parameters. This ensures that audio features, such as Mel-Frequency Cepstral Coefficients (MFCCs) [34], are extracted from high-quality inputs, improving the overall robustness of the framework.

The use of DDPMs enables the system to preprocess both visual and audio data, ensuring that the downstream feature extraction and classification stages operate on high-fidelity inputs.

3.6 Pipeline overview

An overview of the proposed framework is shown in Fig. 1, illustrating the integration of CNNs, ViTs, diffusion models, and multimodal fusion for robust deepfake detection.

4 Experiments

This section presents the experimental setup, datasets, evaluation metrics, results, and a detailed comparison with prior state-of-the-art methods. Furthermore, we provide an analysis of parameter impacts and visualizations for clarity.

Input: Video frames $V = \{F_1, F_2, \dots, F_T\}$, Audio signal A

Output: Binary Classification (Real/Fake)

Initialization:

Load pre-trained CNN, ViT, and Diffusion models.

Step 1: Data Preprocessing

for each frame $F_t \in V$ do

1.1 Visual Refinement:

 Preprocess F_t using the diffusion model to denoise and enhance frame quality.

end

Step 2: Visual Feature Extraction

for each frame $F_t \in V$ do

 Extract key regions (lip, eye, and facial regions) using face detection.

 Pass lip, eye, and facial regions through CNN to capture local features.

 Pass full frame F_t through Vision Transformer (ViT) to extract global features.

 Fuse local features (CNN) and global features (ViT) for each frame.

end

Step 3: Audio Feature Extraction

Extract Mel-Frequency Cepstral Coefficients (MFCC) features from audio signal A .

Model the audio using BiLSTM to capture temporal dependencies.

Step 4: Cross-Modal Synchronization Analysis

for each frame F_t do

 Align the extracted audio features with visual features using Dynamic Time Warping (DTW).

 Compute a similarity score S_t using cosine similarity:

$$S_t = \text{cosine_similarity}(F^{\text{audio}}, F^{\text{visual}})$$

if $S_t < \tau$ (threshold) then

 Mark frame t as "Out-of-Sync".

end

end

Step 5: Multimodal Feature Fusion

Concatenate fused visual features (CNN + ViT) and audio features (BiLSTM).

Pass concatenated multimodal features into a fully connected neural network.

Step 6: Classification

Predict whether the video is Real or Fake using the final classification layer.

return Classification result (Real or Fake).

Algorithm 1: Proposed enhanced deepfake detection framework

4.1 Training setup

The experiments were implemented in PyTorch and executed on an NVIDIA RTX 3090 GPU. The training used the Adam optimizer with a learning rate of 10^{-4} and a batch size of 32. Early stopping and a learning rate scheduler ensured stable convergence. We used metrics such as accuracy, precision, recall, F1-score, and the area under the ROC curve (AUC) to evaluate the model's performance.

4.2 Datasets

The evaluation used the following datasets:

- *FakeAVCeleb* A multimodal dataset with balanced demographics and diverse manipulations [11].
- *AV-Deepfake1M* A challenging dataset with extensive audio-visual forgeries [12].
- *TVIL* A dataset capturing real-world complexities in video manipulations [13].
- *LAV-DF* High- and low-quality deepfake videos emphasizing robustness [14].

4.3 Results and comparison

Table 1 compares the accuracy of our framework with the baseline and previous state-of-the-art methods, including the AVSFF framework.

The results indicate that the proposed framework achieves superior performance across all datasets, exceeding the AVSFF framework and other baselines.

4.4 Extended metric comparisons and analysis

Tables 2, 3, 4, 5, 6 present a comprehensive evaluation of the proposed framework compared with the AVSFF method. Metrics such as accuracy, precision, recall, and F1-score are reported for intra-dataset and cross-dataset settings.

Intra-dataset performance The proposed framework consistently outperforms AVSFF across all datasets. For instance, on the FakeAVCeleb dataset (Table 2), the proposed method achieves an accuracy of 0.9987 compared to 0.9973 by AVSFF. Similar trends are observed in precision, recall, and F1-score, showcasing the robustness of our approach in handling manipulations.

Cross-dataset performance Tables 3, 4, and 5 highlight cross-dataset results when the models are trained on AV-Deepfake1M, TVIL, and LAV-DF, respectively. Key findings are:

- *AV-Deepfake1M (Table 3)* The proposed method achieves an F1-score of 0.9826, significantly surpassing AVSFF's 0.9797. This improvement highlights the effectiveness of integrating ViT and Diffusion Models in handling complex manipulations.
- *TVIL (Table 4)* The proposed framework maintains strong generalization across datasets. For instance, it achieves an F1-score of 0.9906 on TVIL, demonstrating substantial improvements over AVSFF's 0.9880.
- *LAV-DF (Table 5)* The proposed method achieves an accuracy of 0.9825 and an F1-score of 0.9826 on AV-Deepfake1M when trained on LAV-DF, surpassing AVSFF by a notable margin.

Performance on combined dataset Table 6 consolidates the performance when trained on a combined dataset. The proposed framework achieves an accuracy of 0.9989, marking a substantial improvement over AVSFF's 0.9890.

Comparison with other methods Table 7 benchmarks the proposed method against state-of-the-art multimodal deepfake detection approaches. With an accuracy of 0.9988, our framework outperforms all existing models, reinforcing its superiority.

4.4.1 Discussion on Table 5 (trained on LAV-DF)

Table 5 provides insights into the model's robustness when trained on LAV-DF. Key observations include:

- *FakeAVCeleb performance* The proposed method achieves a near-perfect accuracy of 0.9987 and an F1-score of 0.9986, significantly surpassing AVSFF's 0.9478 accuracy. This demonstrates the proposed model's ability to transfer learned features effectively to high-quality datasets.
- *AV-Deepfake1M performance* Despite the challenges posed by this dataset, the proposed method achieves an F1-score of 0.9826, reflecting superior handling of complex manipulations compared to AVSFF's 0.9565.

The results underline the adaptability and generalization capabilities of the proposed framework, particularly in cross-dataset evaluations.

4.5 Limitations

While the proposed framework demonstrates significant improvements in deepfake detection, certain aspects warrant further exploration. The integration of Vision Transformers (ViTs) and diffusion models, although effective in capturing global dependencies and refining noisy inputs, introduces increased computational complexity, potentially impacting real-time applicability. Additionally, the reliance on high-quality training data may limit the generalization capability when applied to low-resolution or highly compressed videos, which are prevalent in real-world scenarios. Furthermore, despite the framework's robustness against existing manipulation techniques, its adaptability to unseen deepfake generation methods remains an open challenge, necessitating further evaluation on diverse datasets. The incorporation of diffusion models as pre-processing steps, while beneficial for feature enhancement, also adds an additional layer of computational overhead, which may not be suitable for deployment in resource-constrained environments. Addressing these challenges in future work could further enhance the efficiency and generalizability of the proposed approach.

Table 1 Comparison of accuracy across datasets using different configurations

Configuration	FakeAVCeleb	AV-Deepfake1M	TVIL	LAV-DF
CNN Only	0.9921	0.9702	0.9810	0.9705
CNN + ViT	0.9963	0.9775	0.9872	0.9789
CNN + ViT + Diffusion	0.9987	0.9825	0.9915	0.9812
AVSFF (Previous work)	0.9973	0.9760	0.9890	0.9786
DeepfakeStack	0.9942	–	0.9805	–
LipForensics	0.9960	0.9755	–	0.9710

Bold values indicate higher is the better

Table 2 Intra- and cross-dataset results (trained on FakeAVCeleb)

Dataset	Method	Accuracy	Precision	Recall	F1-score
FakeAVCeleb	AVSFF	0.9973	0.9971	0.9944	0.9907
FakeAVCeleb	Proposed	0.9987	0.9985	0.9987	0.9986
AV-Deepfake1M	AVSFF	0.9700	0.9680	0.9650	0.9665
AV-Deepfake1M	Proposed	0.9825	0.9832	0.9820	0.9826
TVIL	AVSFF	0.9850	0.9840	0.9825	0.9832
TVIL	Proposed	0.9915	0.9902	0.9910	0.9906
LAV-DF	AVSFF	0.9650	0.9635	0.9600	0.9617
LAV-DF	Proposed	0.9812	0.9824	0.9818	0.9821

Bold values indicate higher is the better

Table 3 Intra- and cross-dataset results (trained on AV-Deepfake1M)

Dataset	Method	Accuracy	Precision	Recall	F1-score
FakeAVCeleb	AVSFF	0.9673	0.9671	0.9644	0.9607
FakeAVCeleb	Proposed	0.9825	0.9812	0.9824	0.9818
AV-Deepfake1M	AVSFF	0.9760	0.9755	0.9780	0.9797
AV-Deepfake1M	Proposed	0.9825	0.9832	0.9820	0.9826
TVIL	AVSFF	0.9400	0.9485	0.9460	0.9472
TVIL	Proposed	0.9789	0.9795	0.9790	0.9792
LAV-DF	AVSFF	0.9500	0.9480	0.9450	0.9465
LAV-DF	Proposed	0.9812	0.9824	0.9818	0.9821

Bold values indicate higher is the better

Table 4 Intra- and cross-dataset results (trained on TVIL)

Dataset	Method	Accuracy	Precision	Recall	F1-score
FakeAVCeleb	AVSFF	0.9573	0.9571	0.9544	0.9507
FakeAVCeleb	Proposed	0.9915	0.9902	0.9910	0.9906
AV-Deepfake1M	AVSFF	0.9450	0.9430	0.9400	0.9415
AV-Deepfake1M	Proposed	0.9789	0.9795	0.9790	0.9792
TVIL	AVSFF	0.9890	0.9897	0.9983	0.9880
TVIL	Proposed	0.9915	0.9902	0.9910	0.9906
LAV-DF	AVSFF	0.9300	0.9280	0.9250	0.9265
LAV-DF	Proposed	0.9812	0.9824	0.9818	0.9821

Bold values indicate higher is the better

Table 5 Intra- and cross-dataset results (trained on LAV-DF)

Dataset	Method	Accuracy	Precision	Recall	F1-score
FakeAVCeleb	AVSFF	0.9478	0.9488	0.9444	0.9407
FakeAVCeleb	Proposed	0.9987	0.9985	0.9987	0.9986
AV-Deepfake1M	AVSFF	0.9600	0.9580	0.9550	0.9565
AV-Deepfake1M	Proposed	0.9825	0.9832	0.9820	0.9826

Bold values indicate higher is the better

Table 6 Intra- and cross-dataset results (trained on combined dataset)

Dataset	Method	Accuracy	Precision	Recall	F1-score
FakeAVCeleb	AVSFF	0.9890	0.9885	0.9870	0.9877
FakeAVCeleb	Proposed	0.9989	0.9990	0.9988	0.9989

Bold values indicate higher is the better

Table 7 Comparison of our method with published multimodality deepfake video detection approaches trained and tested on the various multimodal datasets

References	Model	Method	Modality	Accuracy
[22]	MIS-AVoIDD	Multimodal	AV	0.973
[4]	LipForensics	Unimodal	V	0.76
[10]	Unsupervised multimodal deepfake detection	Multimodal	AV	0.968
[32]	Multimodal-1	Multimodal	AV	0.76
[32]	Multimodal-2	Multimodal	AV	0.674
[32]	CDCN	Multimodal	AV	0.515
[35]	Multimodal alignment	VFD	AV	0.85
[29]	Ensemble + Multimodal	Multimodal	AV	0.89
[36]	Multimodaltrace	Multimodal	AV	0.929
[37]	Novel smart deepfake detection	Multimodal	AV	0.954
	Ours	Multimodal	AV	0.9988

Bold values indicate higher is the better

5 Conclusion

Deepfake detection is a growing challenge due to advanced generative techniques. We propose an enhanced multimodal framework combining local and global features for robust detection, using visual cues (e.g., eye movement, facial regions) and audio. Vision Transformers (ViTs) capture global context, while diffusion models improve preprocessing and feature representation. Our framework outperforms existing methods, achieving state-of-the-art accuracy on the FakeAVCeleb, AV-Deepfake1M, TVIL, and LAV-DF datasets. Ablation studies highlight the importance of ViTs and diffusion models, with pretrained ImageNet weights boosting performance. Compared to frameworks like AVSFF, our approach excels in precision, recall, and F1-score.

This framework offers a scalable solution for deepfake detection, with future work focusing on adapting it to new techniques and real-time performance.

Author Contributions M.J. and T.K. conceptualized the study and wrote the main manuscript text. Z.Z. and F.H.D. designed the experiments and contributed to the development of the multimodal framework. M.J.

prepared and processed the datasets used in the experiments. M.J. and Z.Z. performed the analysis and contributed to the result interpretation. T.K. and F.H.D. prepared the figures. All authors reviewed and approved the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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